

# Water Usage Monitoring System

## Feature Documentation

Resident Portal and Community Admin Portal: what was added, why, how each feature works, and how the complete workflow connects to the existing system.

Stack: React 19 + Vite, Recharts, Spring Boot 3.2 (Java 21), MySQL 8, JWT security

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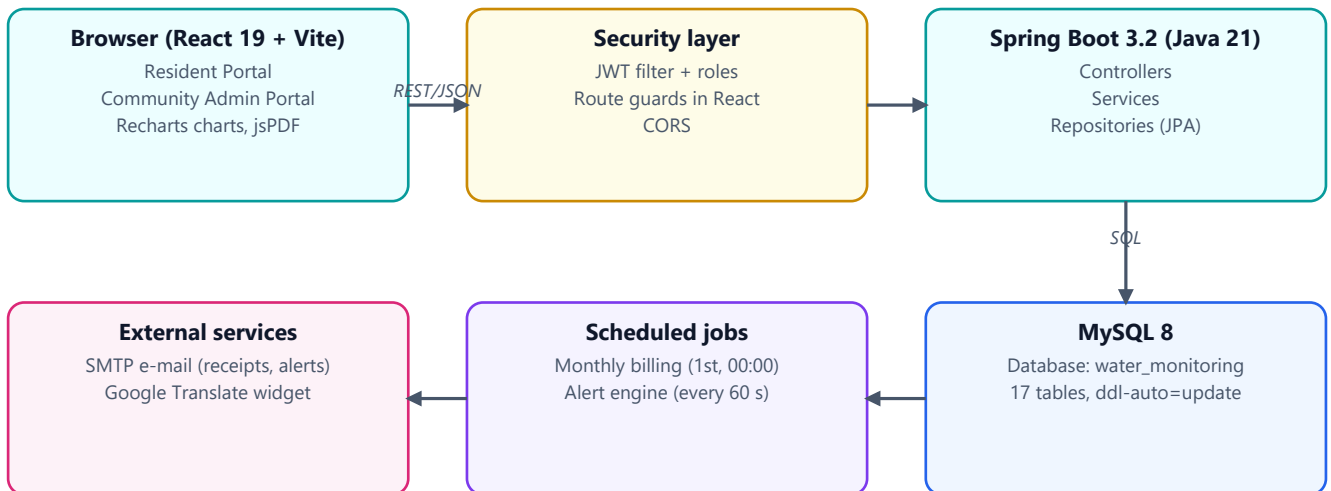
Scope: features as implemented in the code base at the time of writing

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# 1. System overview

The Water Usage Monitoring System lets residents of an apartment community see how much water they use and what they owe, and lets community administrators manage meters, readings, tariffs, billing cycles and bulk water purchases. This document describes the features added to the **Resident Portal** and the **Community Admin Portal**, how each one works, and how they connect to the parts of the system that already existed (billing engine, alert engine, payments).



*Every request carries the JWT; the server derives the household from the login, never from the URL.*

## 1.1 Two portals, two admin interfaces

The system has one resident interface and **two** administrator interfaces. Both administrator interfaces call the same backend.

| Interface                   | URL prefix         | Look                       | Signs in as                     | Purpose                                                                      |
|-----------------------------|--------------------|----------------------------|---------------------------------|------------------------------------------------------------------------------|
| Resident Portal             | /resident/*        | AquaMonitor (dark sidebar) | resident@watermonitor.com       | Household usage, comparison, bills, payments, alerts.                        |
| Community Admin Portal      | /admin/*           | WaterFlow (dark sidebar)   | communityadmin@watermonitor.com | Main administration portal: all new data features in section 4 live here.    |
| Community Admin (Aqua Plus) | /community-admin/* | Aqua Plus (dark sidebar)   | admin@watermonitor.com          | Second admin interface with 11 pages; kept, and given the same header tools. |

**Naming note.** The dashboard titled *Community Admin Dashboard* with the *WaterFlow / Community Admin Portal* sidebar is the **/admin** interface. In this document, "Community Admin Portal" means that interface unless stated otherwise; the Aqua Plus interface is called "Community Admin (Aqua Plus)".

## 1.2 Roles, sign-in and where each login lands

| Login (demo)                            | Server role          | Client role     | Lands on                   | Resident API  | Admin API |
|-----------------------------------------|----------------------|-----------------|----------------------------|---------------|-----------|
| resident@watermonitor.com               | ROLE_RESIDENT        | user            | /resident/dashboard        | Yes           | No (403)  |
| communityadmin@watermonitor.com         | ROLE_PROPERTY_ADMIN  | admin           | /admin/dashboard           | Yes           | Yes       |
| admin@watermonitor.com                  | ROLE_COMMUNITY_ADMIN | community_admin | /community-admin/dashboard | Yes (preview) | Yes       |
| community@aquaplus.com                  | ROLE_COMMUNITY_ADMIN | community_admin | /community-admin/dashboard | Yes (preview) | Yes       |
| Any account created with Create Account | ROLE_RESIDENT        | user            | /resident/dashboard        | Yes           | No (403)  |

Demo passwords are set in AuthService and are intentionally not repeated here. The sign-in page also has an offline fallback that lets the front end open without a server session; pages that need data then show a clear "sign in again" notice (section 3.7).

## 1.3 Security rules enforced by the backend

| Endpoint group                                                              | Who may call it                                                                                           |
|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| /api/auth/**, /api/translate/**, /api/test/**                               | Everyone (login, registration, translation helper).                                                       |
| /api/resident/** (dashboard, consumption comparison)                        | ROLE_RESIDENT, ROLE_PROPERTY_ADMIN, ROLE_COMMUNITY_ADMIN. The household is always derived from the login. |
| /api/community-admin/** (all new admin data APIs)                           | ROLE_PROPERTY_ADMIN and ROLE_COMMUNITY_ADMIN only.                                                        |
| POST on /api/tariffs, /api/billing-cycles, /api/bulk-purchases, /api/meters | ROLE_PROPERTY_ADMIN and ROLE_COMMUNITY_ADMIN only (write access to the older endpoints was locked down).  |
| /api/admin/**                                                               | ROLE_PROPERTY_ADMIN.                                                                                      |
| Everything else (older read endpoints such as bills, alerts, invoices)      | Permitted without a role (existing behaviour; see limitations, section 8).                                |

On the front end, route guards add a second layer: a resident-only page redirects an administrator to their own dashboard unless they entered through the **Resident Portal** header button (preview mode, section 4.2).

## 2. What was added at a glance

### 2.1 Resident Portal

| Feature                                                                                                                                                   | Sidebar location                                                    | Status                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------|
| Resident Dashboard (rebuilt): daily and monthly trend charts, current billing-cycle summary, invoice history with PDF, alerts feed, water-saving tips     | Dashboard > Overview                                                | Replaced the old static page; real data |
| <b>Consumption Comparison</b> (new page): you vs neighbours vs community, daily and monthly, difference, % above/below average, insights, recommendations | Water Management > <b>Consumption Comparison</b> (new sidebar item) | New; real data                          |
| Create Account (e-mail + password) and automatic receipt e-mail after payment                                                                             | Login page; Billing & Payments > Current Bill                       | New flow on existing pages              |
| Session / permission notice with one-click sign-in                                                                                                        | All API-driven resident pages                                       | New                                     |
| Mobile menu drawer and phone-friendly header                                                                                                              | Whole Resident Portal                                               | New                                     |
| Admin preview banner ("Back to admin portal")                                                                                                             | Floating pill on resident pages                                     | New; only for admins                    |

### 2.2 Community Admin Portal (/admin)

| Feature                                                                                                                         | Sidebar location                                 | Status                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------|
| Community Admin Dashboard: live KPIs, supply vs consumption vs billing chart, household chart, billing cycle and tariff, alerts | Dashboard > Overview                             | Rebuilt; real data                                                     |
| Household-wise water-usage comparison charts                                                                                    | Water Management > <b>Household Comparison</b>   | New; real data                                                         |
| Water-meter reading entry, CSV upload and reading history                                                                       | Water Management > <b>Meter Readings</b>         | New; real data                                                         |
| Billing-cycle management                                                                                                        | Billing & Payments > <b>Billing Cycles</b>       | New; real data                                                         |
| Tariff management with version history (never overwritten)                                                                      | Billing & Payments > <b>Tariff Versions</b>      | New; real data                                                         |
| Bulk-water purchase entry and history                                                                                           | Billing & Payments > <b>Bulk Water Purchases</b> | New; real data                                                         |
| Header tools on every page: <b>Resident Portal</b> button/dropdown and Language Selector                                        | Top header of every admin page                   | New                                                                    |
| Consumption Monitoring and Usage Analytics pages                                                                                | (were under Water Management)                    | <b>Removed</b> (fake data); old links redirect to Household Comparison |
| Consistent header, one font stack, mobile drawer, table overflow fixes                                                          | Whole portal                                     | New / fixed                                                            |

## 3. Resident Portal

### 3.1 Resident sidebar (every name)

The resident sidebar has five sections. The only item added is **Consumption Comparison**.

| Section                 | Items                         | Route                            | New?                            |
|-------------------------|-------------------------------|----------------------------------|---------------------------------|
| Dashboard               | Overview                      | /resident/dashboard              | Rebuilt                         |
| Water Management        | Water Consumption             | /resident/water-consumption      | Existing                        |
|                         | Usage History                 | /resident/usage-history          | Existing                        |
|                         | Meter Details                 | /resident/meter-details          | Existing                        |
|                         | <b>Consumption Comparison</b> | /resident/consumption-comparison | <b>New</b>                      |
| Billing & Payments      | Current Bill                  | /resident/current-bill           | Existing (receipt flow updated) |
|                         | Billing History               | /resident/billing-history        | Existing                        |
|                         | Payment History               | /resident/payment-history        | Existing                        |
| Account & Settings      | My Profile                    | /resident/profile                | Existing                        |
|                         | Account Settings              | /resident/settings               | Existing                        |
| Notifications & Support | Notifications                 | /resident/notifications          | Existing                        |

Other resident pages that exist but are reached from inside the portal: Alerts (/resident/alerts) and Help & Support (/resident/help). The sidebar collapses into a slide-in drawer (menu button, top left) below 900 px width. The header always contains the Language Selector and the apartment card (the card is hidden on very small phones to keep the header inside the screen).

### 3.2 Resident Dashboard (rebuilt with real data)

|                         |                                                                                                                                                                                               |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | The landing page of the Resident Portal. It shows the resident's water usage trends, the current billing cycle, invoice history with downloadable PDFs, an alerts feed and water-saving tips. |
| <b>Why it was added</b> | The previous dashboard was static sample content. Residents need one screen that answers three questions: how much water am I using, what do I owe and when, and is anything wrong.           |
| <b>Where to find it</b> | Sidebar: Dashboard > Overview. Route /resident/dashboard. Backend: ResidentDashboardService, ResidentDashboardController.                                                                     |
| <b>Data source</b>      | water_usage, water_meter, apartment, users, bill, invoice, alerts, tariff_plan (read only).                                                                                                   |

#### How it works

- The page calls GET /api/resident/dashboard. The server finds the household from the JWT e-mail: users row, then the apartment whose resident\_user\_id matches; if the account has no linked apartment (demo accounts), the first apartment is used.
- Daily chart:** last 30 days from water\_usage (through the household's meter), area chart in KL per day. Days without a reading are left as gaps, not zeros. A dashed line marks the 30-day average.
- Monthly chart:** last 6 calendar months, summed per month, bar chart. If the household has no readings at all, the chart falls back to billed consumption and says so.
- KPIs:** month-to-date KL and % change against the *same number of days last month* (like-for-like); 30-day daily average (total divided by days that have readings); last month's full total; outstanding amount and number of unpaid bills.
- Billing-cycle card:** the oldest PENDING bill (bills must be paid oldest first); if none, the latest bill. Shows bill number, month, amount due (0 when already paid), usage, bill total, due date and days left or overdue, and status. A Pay button opens Current Bill.

- **Invoice history:** last 12 bills including SUPERSEDED ones; the page lists 6. A PDF button appears only for PAID bills; it downloads the stored invoice through `invoiceService.openInvoice`. Unpaid bills show "Pay to get invoice".
- **Alerts feed:** latest alerts for the apartment (8 returned, 5 shown) with severity dots and relative times.
- **Tips:** contextual tips first, then three general tips that rotate daily (section 3.3 lists the rules).

### 3.3 Water-saving tip rules

| Condition (computed on the server)                                                                     | Tip shown                                                                                                                                                                            | Priority |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Month-to-date use is 15% or more above the same days last month                                        | "Usage is up X% vs last month" plus a leak check (food-colour cistern test)                                                                                                          | High     |
| Month-to-date use is 10% or more below the same days last month                                        | "Nice, usage is down X%"                                                                                                                                                             | Good     |
| From day 5 of the month, projected month (MTD / day of month x days in month) exceeds the Tier 2 limit | "On track for Tier 3 pricing" with the rate and the KL to cut                                                                                                                        | High     |
| Projected month exceeds the Tier 1 limit only                                                          | "Projected N KL" with the Tier 2 rate                                                                                                                                                | Medium   |
| Always                                                                                                 | Three general tips from a pool of eight (fix leaks, shorter showers, tap off, full loads, water plants at dawn, reuse RO reject, bucket not hose, aerators), rotating by day of year | Info     |

### 3.4 Consumption Comparison (new)

|                         |                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | A page that compares the resident's water use with neighbours in the same building and with the whole community, day by day and month by month, and turns the result into plain-language insights and recommendations. |
| <b>Why it was added</b> | A bare number ("21.8 KL") does not tell a resident whether that is high. Comparison gives context, shows the size of any gap in KL and percent, and points to what to do about it.                                     |
| <b>Where to find it</b> | Sidebar: Water Management > Consumption Comparison. Route <code>/resident/consumption-comparison</code> . Backend: <code>ResidentComparisonService</code> , <code>ResidentComparisonController</code> .                |
| <b>Data source</b>      | <code>water_usage</code> , <code>water_meter</code> , apartment (grouping by building), <code>tariff_plan</code> (rate estimate).                                                                                      |

#### How it works

- The page calls `GET /api/resident/consumption-comparison`. Readings of the last six months are grouped by apartment (meter to apartment).
- **Peer groups:** *neighbours* = other households with readings in the same building (block); *community* = all other households with readings. The schema has no apartment size or occupant count, so "similar households" is defined by building; the page states this.
- **Privacy rule:** another household is never shown by name or value. An average is only published if at least **3** other households contributed to it (checked for each day, each month and each period). The building average is also withheld if it could be subtracted from the community average to isolate a few households (it is released only when the rest of the community is empty or at least 3).
- **Daily / Monthly toggle:** daily = last 30 days; monthly = last 6 months (current month is month-to-date for everybody). Series: You, Neighbours, Community average, and your own average (dashed).
- **Summary cards** (for last 30 days or this month so far): your total; difference vs neighbours in KL and % (you minus average, divided by average); the same vs community; and **rank** = the share of other households that use more than you.
- **Difference chart:** one bar per day or month, red when you are above the reference, green when below. The reference is neighbours, else community, else your own average.
- **Interactivity:** hover tooltips show every series and the difference; clicking a legend chip hides or shows a line.

- **Insights** (server rules): 10% or more below the reference is "better than average"; within 10% is "in line"; above 10% is a warning and above 25% is high; last 7 days versus the 23 days before (+15% rising, -10% falling); weekend versus weekday average when weekends are 1.25x or more; days above twice your 30-day average are flagged as possible leaks; with no reference available, you are compared with your own previous 30 days.
- **Recommendations** (up to five): target daily use = reference / 30; excess KL per month = your 30-day total minus the reference; shorter showers and aerators (estimated at 10% of your use); fix drips and running toilets; weekend chore planning; leak check after spikes; "keep it up" when you are well below.
- **Rupee estimate**: excess KL x the marginal rate for your usage level (Tier 1 rate if your 30-day total is within the Tier 1 limit, Tier 2 within the Tier 2 limit, otherwise Tier 3) from the latest tariff. It is labelled an estimate.
- **Not enough data**: with fewer than 3 other households the cards show a dash, a notice explains why, and the charts compare you with your own average; the insights use your previous 30 days.

### 3.5 Create Account and receipt e-mail

|                         |                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Create Account on the sign-in page collects only an e-mail and a password. The e-mail is stored in the database. After a successful payment, the PDF invoice is e-mailed automatically to that same e-mail. |
| <b>Why it was added</b> | Residents should not have to type an e-mail again at payment time, and the receipt must go to the address the account was created with.                                                                     |
| <b>Where to find it</b> | Login page (Create Account); Billing & Payments > Current Bill and Billing History; invoice PDFs from Payment History and the dashboard.                                                                    |
| <b>Data source</b>      | users, roles, user_roles, bill, payment, invoice; SMTP for delivery.                                                                                                                                        |

#### How it works

- The Create Account link sits below the sign-in fields. Registration sends POST /api/auth/register with e-mail, password (8 characters or more) and role ROLE\_RESIDENT. The e-mail is trimmed and lower-cased; the display name is derived from the part before the @.
- After creation, the form returns to sign-in with the e-mail filled in and a success message.
- Payment: POST /api/bills/{id}/pay. The controller reads the authenticated e-mail from the JWT and passes it to PaymentService; the payment screen no longer asks for a Gmail address.
- PaymentService rules: a PAID bill cannot be paid again; a SUPERSEDED bill cannot be paid; older PENDING bills must be paid first (the error names the oldest bill).
- On success the bill becomes PAID, a Payment row is written, an invoice PDF is generated and saved in invoice, and the e-mail with the PDF is sent to the account e-mail (falling back to the household resident's e-mail). E-mail sending is best effort: a mail failure never undoes the payment.

### 3.6 Bill and calculation rules the resident sees

Bills are computed by TariffCalculator (shared by bill generation and by recalculation after a tariff change):

```

volumetric = tier1 part + tier2 part + tier3 part
tier1 part = min(use, T1 limit) x T1 rate
tier2 part = (min(use, T2 limit) - T1 limit) x T2 rate      (if use > T1 limit)
tier3 part = (use - T2 limit) x T3 rate                    (if use > T2 limit)
subtotal   = volumetric + fixed base charge + (common water charge + bulk share)
GST         = 18% of subtotal
total      = subtotal + GST

```

**Worked example with the current tariff (version 9):** limits 10 KL and 20 KL, rates 25, 75.5 and 42.5 per KL, base 180, common 320. For 25 KL: 10 x 25 = 250, 10 x 75.5 = 755, 5 x 42.5 = 212.5, so volumetric = 1,217.50. Subtotal = 1,217.50 + 180 + 320 = 1,717.50. GST = 309.15. Total = **Rs 2,026.65**. This matches bill BILL-98C5E753 in the database.

**Bill status:** PENDING (unpaid), PAID, or SUPERSEDED (replaced by a recalculated bill after a tariff change; kept for audit, never payable).

### 3.7 Session notice and permission handling

When a resident API call fails, the page shows a clear message instead of a generic error. An HTTP 401/403 means the login cannot use resident data (an admin account, an expired session, or an offline login with no token). The notice names the account, and a **Sign in as a resident** button clears the session and opens the login page. Other failures show a Retry button. The shared API helper now keeps the HTTP status on errors and tolerates empty response bodies (the security layer returns an empty 403).

### 3.8 Responsiveness, font and language

- **One font** across the whole application: the Inter stack defined once as a CSS variable; form controls inherit it; monospace IDs use Inter with tabular digits.
- **Language Selector**: available in the header of every resident page (and the login page). It stores the language, sets the translation cookie, and reloads so the Google Translate widget can translate the page.
- **Phone layout**: sidebar drawer, header without the apartment card, single-column cards, charts that resize with their container.

## 4. Community Admin Portal (/admin)

### 4.1 Sidebar (every name)

Seven expandable sections. The active page's section opens automatically. Items marked NEW were added; items marked REMOVED no longer exist.

| Section               | Sidebar item                | Route                                | Status         | Data                               |
|-----------------------|-----------------------------|--------------------------------------|----------------|------------------------------------|
| Dashboard             | Overview                    | /admin/dashboard                     | Rebuilt        | Real (MySQL)                       |
| User Management       | Residents                   | /admin/residents                     | Existing       | Sample content                     |
|                       | Add Resident                | /admin/add-resident                  | Existing       | API (resident service)             |
|                       | User Accounts               | /admin/user-accounts                 | Existing       | Sample content                     |
| Water Management      | Water Meters                | /admin/water-management/water-meters | Existing       | API (meters); may show demo meters |
|                       | Add Water Meter             | /admin/add-water-meter               | Existing       | API                                |
|                       | <b>Household Comparison</b> | /admin/household-comparison          | <b>NEW</b>     | Real                               |
|                       | <b>Meter Readings</b>       | /admin/meter-readings                | <b>NEW</b>     | Real                               |
|                       | Consumption Monitoring      | (redirects to Household Comparison)  | <b>REMOVED</b> | Was simulated                      |
|                       | Usage Analytics             | (redirects to Household Comparison)  | <b>REMOVED</b> | Was hard-coded                     |
| Billing & Payments    | Billing Management          | /admin/billing-management            | Existing       | API (bills, bulk purchases)        |
|                       | Payment Management          | /admin/payment-management            | Existing       | Sample content                     |
|                       | Tariff Management           | /admin/tariff-management             | Existing       | API (tariff, bill generation)      |
|                       | <b>Tariff Versions</b>      | /admin/tariff-versions               | <b>NEW</b>     | Real                               |
|                       | <b>Billing Cycles</b>       | /admin/billing-cycles                | <b>NEW</b>     | Real                               |
|                       | <b>Bulk Water Purchases</b> | /admin/bulk-purchases                | <b>NEW</b>     | Real                               |
|                       | Traffic Analytics           | /admin/traffic-management            | Existing       | Sample content                     |
| Operations            | Meter Monitoring            | /admin/meter-monitoring              | Existing       | Sample content                     |
|                       | Alerts & Notifications      | /admin/alerts-notifications          | Existing       | API (alert engine)                 |
| Service & Maintenance | Service Requests            | /admin/service-requests              | Existing       | Sample content                     |
|                       | Maintenance Management      | /admin/maintenance-management        | Existing       | Sample content                     |
|                       | Issue & Resolution Tracking | /admin/issue-tracking                | Existing       | Sample content                     |
| Reports & Analytics   | Consumption Reports         | /admin/consumption-reports           | Existing       | Sample content                     |
|                       | Billing Reports             | /admin/billing-reports               | Existing       | Sample content                     |
|                       | Revenue Reports             | /admin/revenue-reports               | Existing       | Sample content                     |
| System                | Admin Profile               | /admin/profile                       | Existing       | Sample content                     |

System Settings (/admin/settings) also exists as a page but is not in the sidebar. The profile card in the sidebar shows the signed-in e-mail.

## 4.1b Community Admin (Aqua Plus) sidebar (every name)

The second admin interface (/community-admin/\*) has one page per sidebar entry. All eleven pages contain built-in sample content and no API calls; they were kept and received the same header tools and phone layout.

| Sidebar section / item | Route                                   |
|------------------------|-----------------------------------------|
| Dashboard > Overview   | /community-admin/dashboard              |
| Residents              | /community-admin/residents              |
| Admin Management       | /community-admin/admin-management       |
| User Accounts          | /community-admin/user-accounts          |
| Water Meters           | /community-admin/water-meters           |
| Consumption Monitoring | /community-admin/consumption-monitoring |
| Billing & Payments     | /community-admin/billing-payments       |
| Tariff Management      | /community-admin/tariff-management      |
| Alerts & Notifications | /community-admin/alerts-notifications   |
| Reports                | /community-admin/reports                |
| Settings               | /community-admin/settings               |

## 4.2 Header tools: Resident Portal button and Language Selector

|                         |                                                                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Every admin page (both interfaces) has, at the top right of its header, a small <b>Resident Portal</b> split button and the <b>Language Selector</b> . |
| <b>Why it was added</b> | Administrators need to see what residents see, and every portal must offer the same language control.                                                  |
| <b>Where to find it</b> | Top header of every /admin/* and /community-admin/* page.                                                                                              |
| <b>Data source</b>      | No new tables. Uses browser storage for the language and the preview flag.                                                                             |

### How it works

- The main button opens the Resident Portal dashboard directly. The small arrow opens a menu: Dashboard, Consumption Comparison, Usage History, Current Bill, Billing History, Alerts.
- Clicking sets a per-tab preview flag (sessionStorage). Resident-only routes accept an administrator only when this flag is set. A floating pill then reads "Previewing the Resident Portal as an admin" with a **Back to admin portal** button that clears the flag and returns to the right admin dashboard.
- The backend now also accepts ROLE\_COMMUNITY\_ADMIN on /api/resident/\*\*, so the preview can load data for both admin roles. An administrator sees the demo household (the first apartment), not a real resident's private data.
- Implementation: one component, PortalTools, is mounted by each admin sidebar. It inserts the tools into whichever header the page renders (and re-inserts them if a page swaps its header). This puts the same controls in the same place on every page without editing dozens of page files.
- The Language Selector is the same component used by the Resident Portal. The chosen language is stored, so it stays selected when moving between the Resident, Admin and Community Admin portals.

## 4.3 Community Admin Dashboard (rebuilt)

|                         |                                                                                                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | The admin landing page. Shortcut cards to the new modules, four KPI cards, a six-month supply/consumption/billing chart, the open billing cycle and current tariff, a household usage chart, and the latest system alerts. |
| <b>Why it was added</b> | The old dashboard showed fixed numbers (for example 142 households). Administrators need real figures from the database.                                                                                                   |
| <b>Where to find it</b> | Sidebar: Dashboard > Overview. Route /admin/dashboard.                                                                                                                                                                     |
| <b>Data source</b>      | apartment, water_meter, water_usage, bill, billing_cycle, bulk_water_purchase, tariff_plan, alerts.                                                                                                                        |

### How it works

- Calls GET /api/community-admin/dashboard (roles PROPERTY\_ADMIN or COMMUNITY\_ADMIN).
- **Households**: count of apartments, and how many are OCCUPIED. **Smart meters**: total, ONLINE, and the rest (need attention).
- **Consumption this month**: total of all readings dated in the current month; **average per day** = that total / day of month.
- **Revenue collected**: sum of PAID bills (SUPERSEDED bills are excluded); **outstanding**: sum of PENDING bills; paid and pending counts; **collection rate** = collected / (collected + outstanding) x 100.
- **Six-month chart**: bars for bulk water purchased (KL, by delivery month) and water consumed (KL, sum of readings), plus a line for billed rupees (bills created that month). Two axes: KL on the left, rupees on the right. Comparing purchased with consumed shows possible network loss.
- **Billing cycle and tariff card**: the OPEN cycle with its consumption, bulk volume and bills, and the latest tariff version with its tier rates.
- **Household chart**: the top 8 households this month against last month, with the community average as a dashed line.
- **Alerts**: the six most recent alerts across the community with severity.

## 4.4 Household Comparison (household-wise usage charts)

|                         |                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Compares every household's consumption for a chosen month against the previous month and the community average. |
| <b>Why it was added</b> | Administrators need to spot high users, unusual changes and the spread of consumption across households.        |
| <b>Where to find it</b> | Sidebar: Water Management > Household Comparison. Route /admin/household-comparison.                            |
| <b>Data source</b>      | apartment, water_meter, water_usage.                                                                            |

### How it works

- Calls GET /api/community-admin/households/usage?month=YYYY-MM (the month picker defaults to the current month; an invalid month returns a 400 message).
- For each apartment: consumption in the month (through its meter), the previous month, change %, share % of the community total, and the number of days with readings. Rows are sorted highest first (or by apartment on the page).
- For the current month the previous month is limited to the same number of days, so the comparison is fair.
- **Average per household** = community total / number of apartments. **High users** = households above 1.5 x the average.
- **Chart**: horizontal grouped bars (previous month grey, this month teal) per household, with a dashed average line; height grows with the number of households. A detail table repeats the numbers with a change arrow (red up, green down).

## 4.5 Meter Readings (upload and management)

|                         |                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Enter a reading for a meter, upload many readings in a CSV file, and review the reading history. It also lists all registered meters with battery, signal, status and the last reading. |
| <b>Why it was added</b> | The system's calculations, charts and alerts all depend on water_usage; administrators needed a validated way to put readings in.                                                       |
| <b>Where to find it</b> | Sidebar: Water Management > Meter Readings. Route /admin/meter-readings.                                                                                                                |
| <b>Data source</b>      | water_meter (read, status update), water_usage (insert).                                                                                                                                |

### How it works

- **Single entry** (POST /api/community-admin/meters/{id}/readings): meter, reading date, current reading, optional previous reading. If the previous reading is left blank it is taken from the meter's latest reading. Consumption = current minus previous, rounded to two decimals.
- **Validation** (server side, repeated in the form): meter must exist; date required and not in the future; current reading between 0 and 10,000,000; no second reading for the same meter and date; previous reading not negative; current must not be lower than previous.
- **Leak flag**: a single reading above 5.0 KL sets the meter status to ALERT, and the alert engine evaluates the new reading on its next run.
- **CSV upload** (POST /api/community-admin/readings/upload): columns meter, current, date (YYYY-MM-DD), previous (optional); the meter can be its serial number or id; a header row is optional; only .csv files (5 MB limit in the page). Each row goes through the same rules as single entry. Invalid rows are skipped and reported with their line number, while valid rows are still imported (the result shows imported, rejected and total rows).
- **History**: the most recent 30 readings, filterable by meter, with High marked above 5 KL. Arriving from Water Meters with ?serial= preselects that meter.

## 4.6 Billing Cycles

|                         |                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Create billing periods, see what each period produced, and close the open period.                                                    |
| <b>Why it was added</b> | Bills, readings and bulk purchases all belong to a period. Administrators needed a controlled way to define and close those periods. |
| <b>Where to find it</b> | Sidebar: Billing & Payments > Billing Cycles. Route /admin/billing-cycles.                                                           |
| <b>Data source</b>      | billing_cycle (insert, status update); reads water_usage, bill, bulk_water_purchase.                                                 |

### How it works

- **Create** (POST /api/community-admin/cycles): name (required, 50 characters at most, unique), start date and end date. The end must be after the start, the length cannot exceed about a year, and the dates cannot overlap any existing cycle.
- **Status**: if no cycle is OPEN the new one becomes OPEN, otherwise UPCOMING. Only one cycle is OPEN at a time.
- **Close** (POST /cycles/{id}/close): only the OPEN cycle can be closed; it becomes CLOSED and the earliest UPCOMING cycle (if any) becomes OPEN.
- **Per-cycle figures**: consumption = readings dated within the cycle; bulk purchased = purchases linked to the cycle; bills and billed amount = non-superseded bills created within the cycle dates.
- The page shows the open cycle, its consumption and billed amount, total and closed counts, and a table with a Close button on the open row (with a confirmation).

## 4.7 Tariff Versions (version history, never overwritten)

|                         |                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Shows the complete history of tariff versions and lets an administrator save new rates as a new version.                    |
| <b>Why it was added</b> | A tariff change must never destroy the past: bills, audits and disputes need to know which rates applied when.              |
| <b>Where to find it</b> | Sidebar: Billing & Payments > Tariff Versions. Route /admin/tariff-versions. The older Tariff Management page still exists. |
| <b>Data source</b>      | tariff_plan (insert only).                                                                                                  |

### How it works

- **View** (GET /api/community-admin/tariffs): the current tariff and every version, newest first. Values that changed compared with the version before are highlighted.
- **Save** (POST /api/community-admin/tariffs): the server *inserts a new row* with version = highest version + 1 and effective-from = now. The previous rows are never modified (verified: saving v10 left v9 untouched).
- **Validation:** plan name required (100 characters at most); tier limits 0.1 to 1,000 KL; tier rates 0 to 1,000; base and common charges 0 to 100,000; the Tier 2 limit must be greater than the Tier 1 limit; a save identical to the current tariff is rejected.
- The form is pre-filled from the current version and shows "was X" beside every edited field; saving asks for confirmation.
- Saving a version does not itself change issued bills. Pending bills are recalculated when bills are generated again (the Tariff Management page does this after its save, and the monthly job does it on the 1st); the old bill becomes SUPERSEDED and a new PENDING bill is issued with an alert.

## 4.8 Bulk Water Purchases

|                         |                                                                                                                                                         |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What it is</b>       | Records tanker and municipal water deliveries and shows purchase history with volumes, costs and unit costs.                                            |
| <b>Why it was added</b> | The community buys water in bulk and shares the cost. That cost needs to be recorded so it can feed the bills and be compared with what households use. |
| <b>Where to find it</b> | Sidebar: Billing & Payments > Bulk Water Purchases. Route /admin/bulk-purchases.                                                                        |
| <b>Data source</b>      | bulk_water_purchase (insert); reads billing_cycle.                                                                                                      |

### How it works

- **Entry** (POST /api/community-admin/bulk-purchases): supplier, delivery date, water source (tanker, municipal supply, borewell, other), volume in KL, total cost, optional notes.
- **Validation:** supplier required (100 characters at most); date required, not in the future and not more than 5 years old; volume above 0 and at most 1,000,000 KL; cost above 0; notes up to 2,000 characters.
- **Unit cost** = total cost / volume, calculated by the server (the form previews it).
- **Cycle link:** the purchase is linked to the billing cycle whose dates contain the delivery date, otherwise to the open cycle, otherwise none.
- **Summary:** number of purchases, total volume, total cost and average unit cost across all purchases; a history table lists deliveries newest first.
- **Connection to billing:** the bill generator adds the bulk cost share to each bill's common-water charge (section 5.1).

## 4.9 Monitoring and alerts (existing engine that the new features feed)

The alert engine already existed. The new Meter Readings feature supplies its input, and the new dashboards display its output. The engine runs every 60 seconds, and also immediately after a tariff is saved through the older Tariff Management page.

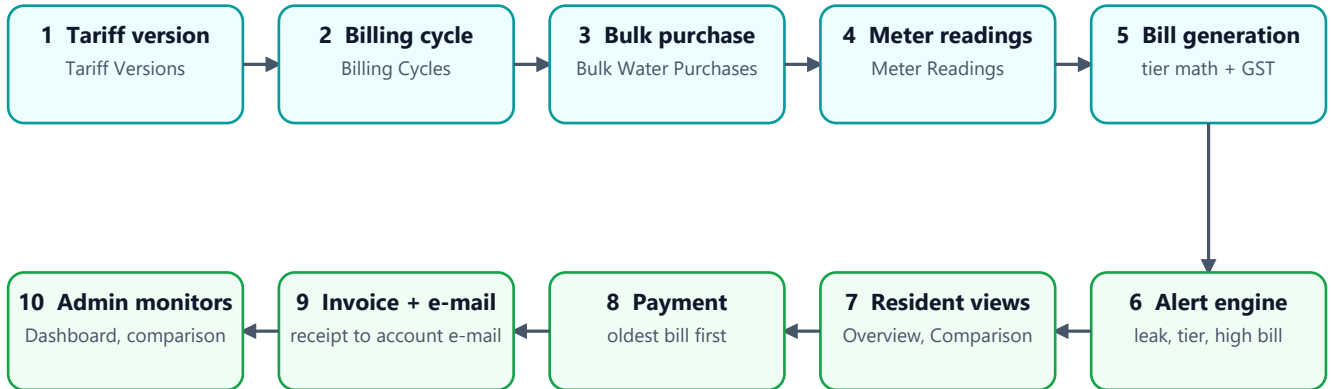
| Alert type                | Rule                                                                                                                          | Notes                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| POTENTIAL_LEAK            | Latest reading is more than mean + k x standard deviation of the meter's earlier readings (k = 2.0 by default) and above 5 KL | Needs at least 2 readings; if the deviation is below 0.5 it is treated as 1.0; one alert per reading. |
| HIGH_CONSUMPTION          | A reading exceeds the configured threshold (25 KL by default)                                                                 | One alert per reading.                                                                                |
| HIGH_BILL / critical bill | A bill's total is more than 1.2x (high) or 1.5x (critical) the amount expected from the tariff for its consumption            | One alert per bill.                                                                                   |
| TIER_ESCALATION           | Consumption in a reading or bill enters Tier 2 or Tier 3; "very critical" when above Tier 2 limit + 10 KL                     | Uses the current tariff limits.                                                                       |
| Recalculation notice      | A pending bill is replaced after a tariff change                                                                              | Explains old vs new amount to the resident.                                                           |

Thresholds (bill overcharge, consumption limit, sigma multiplier, e-mail alerts on/off) are held in AlertConfigDto and can be changed through the alert API. Residents see alerts on Alerts, Notifications and the dashboard feed; administrators see them on Alerts & Notifications and the dashboard's alert list. Meter status feeds the dashboard's "online / need attention" counts.

## 4.10 Removed pages, consistency fixes and phone layout

- **Consumption Monitoring** and **Usage Analytics** were removed. Both showed invented data: fake meters and residents, a "live flow" generated by a random number, fixed monthly bars. The database has no flow-rate data, and Household Comparison, Meter Readings and Water Meters now cover the real parts. Old addresses redirect to Household Comparison, and the Water Meters "Usage" button opens Meter Readings for that meter.
- **Header consistency:** all pages share one white header bar (32 px side padding, 22 px bold title). Older pages had lost their header styling and titles touched the sidebar; a shared style now applies everywhere. The two pages with headings inside the content (Tariff Management, System Settings) were given the same bar.
- **Content under the sidebar** at 900 to 1024 px, a dark unreadable Settings page, and pages wider than the screen (Billing Management, Payment Management, Traffic Analytics) were fixed; wide tables now scroll inside their card.
- **One font:** the whole application uses one font stack (Inter with system fallbacks) defined once.
- **Logout** now clears the login token and e-mail as well, and the sidebar profile shows the signed-in e-mail.
- **Phone layout:** a menu button opens the sidebar as a drawer on every admin page; header tools take their own row; tables scroll inside their card.

## 5. Complete workflow and how it connects to the existing system



*Admin steps (1-4) feed the billing engine; residents see the result and pay; the admin dashboards read it back.*

| #  | Who / where                                           | Action                       | API                                              | Tables                                                  | Result                                                          |
|----|-------------------------------------------------------|------------------------------|--------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------|
| 1  | Admin: Tariff Versions                                | Save new rates               | POST /community-admin/tariffs                    | tariff_plan (insert)                                    | New version becomes current; history kept.                      |
| 2  | Admin: Billing Cycles                                 | Create a cycle               | POST /community-admin/cycles                     | billing_cycle                                           | OPEN or UPCOMING period.                                        |
| 3  | Admin: Bulk Water Purchases                           | Record a delivery            | POST /community-admin/bulk-purchases             | bulk_water_purchase                                     | Unit cost computed; linked to cycle; counts toward bill share.  |
| 4  | Admin: Meter Readings                                 | Enter or upload readings     | POST /meters/{id}/readings, /readings/upload     | water_usage, water_meter                                | Consumption per day; meter set to ALERT above 5 KL.             |
| 5  | System: monthly job or Billing/Tariff Management      | Generate bills               | POST /api/bills/generate-monthly (existing)      | bill                                                    | Tier math + bulk share + GST; changed pending bills superseded. |
| 6  | System: alert engine (every 60 s)                     | Evaluate readings and bills  | internal                                         | alerts                                                  | Leak, threshold, tier and bill alerts; optional e-mail.         |
| 7  | Resident: Overview, Consumption Comparison            | View usage, bill, comparison | GET /resident/dashboard, /consumption-comparison | reads apartment, water_usage, bill, alerts, tariff_plan | Charts, insights, tips, amount due.                             |
| 8  | Resident: Current Bill                                | Pay oldest pending bill      | POST /api/bills/{id}/pay                         | payment, bill                                           | Bill becomes PAID.                                              |
| 9  | System                                                | Invoice and receipt          | internal                                         | invoice                                                 | PDF stored and e-mailed to the account e-mail.                  |
| 10 | Admin: Overview, Household Comparison, Billing Cycles | Monitor                      | GET /community-admin/*                           | reads all of the above                                  | Revenue, collection rate, usage by household, cycle totals.     |

## 5.1 Calculation reference

| Quantity                    | Formula                                                                                                                   | Where used                                   |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Bill amount                 | Tier 1/2/3 volumetric + base + (common + bulk share), plus 18% GST (section 3.6)                                          | Bill generation, recalculation, bill alerts  |
| Bulk share per apartment    | (sum of the cost of all bulk purchases) / (number of apartments), rounded to 2 decimals, added to the common water charge | Bill generation                              |
| Consumption of a reading    | current reading - previous reading (previous defaults to the meter's latest reading)                                      | Meter Readings                               |
| Percent above/below average | (you - average) / average x 100                                                                                           | Consumption Comparison, Household Comparison |
| Average per household       | community total / number of apartments (admin); mean of contributing households (resident comparison)                     | Household Comparison; Resident comparison    |
| Rank                        | share of other households with a higher total than yours                                                                  | Consumption Comparison                       |
| Projected month usage       | month-to-date / day of month x days in month                                                                              | Dashboard tips                               |
| Leak threshold              | mean + 2 x standard deviation of earlier readings (and above 5 KL)                                                        | Alert engine                                 |
| Collection rate             | collected / (collected + outstanding) x 100                                                                               | Admin dashboard                              |
| Saving estimate             | excess KL over the reference x marginal tier rate                                                                         | Consumption Comparison                       |

## 5.2 Bill lifecycle rules

- A new month creates one PENDING bill per apartment. If a PENDING bill for the same month already exists and the recomputed total differs, the old bill is marked SUPERSEDED (kept for audit) and a new PENDING bill is inserted; the resident is notified.
- PENDING bills from an earlier month are left alone; PAID bills are never touched.
- Payment order is oldest first; the payment message names the oldest unpaid bill.

## 6. API and database reference

### 6.1 New endpoints

| Method and path                                   | Purpose                                                 | Roles            |
|---------------------------------------------------|---------------------------------------------------------|------------------|
| GET /api/resident/dashboard                       | Everything the resident dashboard shows                 | Resident, admins |
| GET /api/resident/consumption-comparison          | Comparison series, summaries, insights, recommendations | Resident, admins |
| GET /api/community-admin/dashboard                | Admin KPIs, chart series, cycle, tariff, alerts         | Admins           |
| GET /api/community-admin/households/usage?month=  | Household comparison for a month                        | Admins           |
| GET /api/community-admin/meters                   | Meters with apartment and last reading                  | Admins           |
| GET /api/community-admin/readings?meterId=&limit= | Recent readings (up to 200)                             | Admins           |
| POST /api/community-admin/meters/{id}/readings    | Add one reading (validated)                             | Admins           |
| POST /api/community-admin/readings/upload         | CSV import with per-row errors                          | Admins           |
| GET / POST /api/community-admin/cycles            | List and create billing cycles                          | Admins           |
| POST /api/community-admin/cycles/{id}/close       | Close the open cycle                                    | Admins           |
| GET / POST /api/community-admin/tariffs           | Tariff versions (append only)                           | Admins           |
| GET / POST /api/community-admin/bulk-purchases    | Bulk purchases and summary                              | Admins           |

### 6.2 Tables (MySQL database water\_monitoring)

| Table                                    | Holds                                                                    | Used by the new features for               |
|------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------|
| apartment                                | Households: number, building, floor, occupancy, linked resident user     | Grouping (building), household list        |
| water_meter                              | Meters: serial, apartment, battery, signal, status                       | Meter list, status counts, leak flag       |
| water_usage                              | Readings: meter, previous, current, consumption, date                    | Charts, comparison, dashboards, alerts     |
| bill                                     | Bills: number, month, consumption, amounts, GST, total, due date, status | Billing cycle card, invoices, revenue KPIs |
| payment / invoice                        | Payments and stored invoice PDFs                                         | PDF download, receipt e-mail               |
| tariff_plan                              | Tariff versions with tier limits and rates                               | Tariff Versions, bill math, estimates      |
| billing_cycle                            | Cycles: name, dates, status                                              | Billing Cycles                             |
| bulk_water_purchase                      | Deliveries: supplier, date, volume, cost, unit cost, cycle               | Bulk Water Purchases, bill share           |
| alerts                                   | Alerts with type, severity, status                                       | Alert feeds                                |
| users, roles, user_roles, resident       | Accounts and roles                                                       | Sign-in, account e-mail                    |
| building, complaint, maintenance_request | Existing supporting data                                                 | Not changed by the new features            |

## 7. Which pages use real data

This matrix separates features backed by the database from pages that still contain built-in sample content, so it is clear what is implemented.

| Portal                      | Real data (database)                                                                                                                                                                                              | Built-in sample content (no API calls)                                                                                                                                                                                            |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Resident                    | Overview, Consumption Comparison, Current Bill, Billing History, Payment History, Alerts, Notifications                                                                                                           | Water Consumption, Usage History, Meter Details, My Profile, Account Settings, Help & Support                                                                                                                                     |
| Community Admin (/admin)    | Overview, Household Comparison, Meter Readings, Billing Cycles, Tariff Versions, Bulk Water Purchases, Billing Management, Tariff Management, Alerts & Notifications, Add Resident, Add Water Meter, Water Meters | Residents, User Accounts, Payment Management, Traffic Analytics, Meter Monitoring, Service Requests, Maintenance Management, Issue & Resolution Tracking, Consumption / Billing / Revenue Reports, Admin Profile, System Settings |
| Community Admin (Aqua Plus) | None                                                                                                                                                                                                              | All eleven pages                                                                                                                                                                                                                  |

## 8. Current limitations and points to be aware of

- **Bill consumption is not yet taken from meter readings.** The bill generator uses a consumption value passed to it (25 KL by default) rather than summing each apartment's readings, so bills and readings are separate data today. Connecting them is the natural next step.
- **Bulk cost apportionment is all-time.** The share added to every bill is the total cost of all purchases divided by the number of apartments, not the purchases of the billing cycle.
- **Similar-sized households** are approximated by building because the database has no size or occupant data. With the current data (one household) the comparison page shows the "not enough households" state until at least 3 other households have readings.
- **Older endpoints** (bills, alerts, invoices, and read access to tariffs, cycles, purchases and meters) are still readable without a role, and older write endpoints are role-protected but not field-validated. The new `/api/community-admin` endpoints are fully validated.
- **Sample-content pages** (section 7) still show fixed data.
- **Translation** relies on the Google Translate website widget, so it needs internet access and translates after the page loads.
- **Demo readings:** when the readings table is empty at start-up, 180 days of readings are generated for the first apartment only so the charts are not blank. It never runs once any real reading exists.
- **Preview mode** shows the demo household, not a chosen resident.

**Live snapshot at the time of writing.** Database: 1 apartment, 1 meter, 180 readings (26 March to 21 September 2026), tariff version 9, 14 bills (12 PAID totalling Rs 19,529.73; 2 SUPERSEDED). The resident test account shows 21.8 KL over the last 30 days, 14.76 KL month to date.